

LABORATORY RECORD**EXP NO.: 6 IMPLEMENTING FABRIC CHANNELS FOR PRIVACY****AIM**

To create and configure multiple channels in a Hyperledger Fabric network, deploy chaincode on a specific channel, and demonstrate how Fabric channels provide data privacy and isolation between organizations.

PROCEDURE**Step 1: Navigate to the Fabric Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Verify the Existing Channel

```
peer channel list
```

Step 4: Create a New Private Channel

```
./network.sh createChannel -c privatechannel
```

Step 5: Verify Available Channels

```
peer channel list
```

Step 6: Deploy Chaincode on the Private Channel

```
./network.sh deployCC -c privatechannel -ccn basic \  
-ccl javascript -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 7: Initialize the Ledger

```
peer chaincode invoke -o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com --tls \  
--cafile $ORDERER_CA -C privatechannel -n basic \  
-c '{"function":"InitLedger","Args":[]}'
```

Step 8: Query Assets from the Private Channel

```
peer chaincode query -C privatechannel -n basic \  
-c '{"Args":["GetAllAssets"]}'
```

Step 9: Verify Channel Isolation

```
peer chaincode query -C mychannel -n basic \
  -c '{"Args":["ReadAsset","asset1"]}'
```

Step 10: Stop the Fabric Network

```
./network.sh down
```

PROGRAM

Fabric Channel Creation & Private Chaincode Deployment Shell Script (deploy_private_channel.sh):

```
#!/bin/bash
# Script to create channels and deploy chaincode for data privacy

cd ~/fabric-dev/fabric-samples/test-network

# Start network and create default channel
./network.sh up createChannel -ca

# Create a separate private channel
./network.sh createChannel -c privatechannel

# Verify created channels
peer channel list

# Deploy asset transfer chaincode to privatechannel
./network.sh deployCC -c privatechannel -ccn basic -ccl javascript -ccp ../
asset-transfer-basic/chaincode-javascript

echo "Fabric Channels and Private Chaincode Deployed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
Channels peers has joined:
mychannel
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh createChannel -c
privatechannel
Channel 'privatechannel' created successfully.

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
Channels peers has joined:
mychannel
privatechannel
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -c
privatechannel -ccn basic -ccl javascript -ccp ../asset-transfer-basic/chaincode-
javascript
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode...
Committing Chaincode...
Chaincode committed successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C privatechannel -n basic -c '{"function":"InitLedger","Args":[]}'
Ledger Initialized Successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C
privatechannel -n basic -c '{"Args":["GetAllAssets"]}'
[
  { "ID": "asset1", "Owner": "Tomoko", "Color": "Blue" },
  { "ID": "asset2", "Owner": "Brad", "Color": "Red" }
]

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C
mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
Error: endorsement failure during query. response status 500: chaincode 'basic' not
found on channel 'mychannel'
Data stored in privatechannel is not accessible through mychannel. Channel privacy
verified successfully.

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Network cleaned up successfully
```

RESULT

Hence the experiment was done successfully.